

60 MCQs, 30 short, 20 long, 40 viva, 15 case-based, 15 data-interpretation questions - with the full instructor answer key and rubrics

Instructor assessment package

Instructor: Dr. Saira Azam - Assistant Professor, Centre of Excellence in
Molecular Biology, University of the Punjab, Lahore

All datasets used in this package are simulated teaching data, clearly labelled as such. This course involves no wet-lab work with living GMOs; case studies and regulatory summaries reflect published evidence and should be verified against current primary sources.

Table of Contents

Placeholder for table of contents	0
-----------------------------------	---

MCQs — GMO Environment and Risk Assessments

60 multiple-choice questions · Recall → Understanding → Application → Analysis → Evaluation

Instructor: Dr. Saira Azam, CEMB, University of the Punjab, Lahore. Answer key in Answer-Key.md.

Section A — Recall (Q1–Q12)

Q1. Which term describes the *inherent potential* of an agent to cause harm?

- (a) Risk (b) Exposure (c) Hazard (d) Consequence

Q2. Risk characterization integrates:

- (a) Only hazards (b) Hazard, exposure, dose–response, uncertainty (c) Only exposure (d) Regulation and policy

Q3. The Cartagena Protocol governs:

- (a) Food labelling (b) Transboundary movement of LMOs (c) Pesticide residues (d) Clinical trials

Q4. The first stage of the ERA framework is:

- (a) Hazard identification (b) Risk management (c) Problem formulation (d) Monitoring

Q5. "PMEM" stands for:

- (a) Pre-market evaluation method (b) Post-market environmental monitoring (c) Protein-mediated exposure model (d) Population management and ecology module

Q6. Which agency in the US Coordinated Framework evaluates plant-incorporated protectants (e.g., Bt toxins)?

- (a) FDA (b) USDA-APHIS (c) EPA (d) CDC

Q7. Gene flow requires, at minimum:

- (a) Soil contact (b) Sexual compatibility and hybrid viability (c) Insect vectors (d) Horizontal transfer

Q8. An LC50 is:

- (a) The lowest observed adverse concentration (b) The dose causing 50% mortality (c) A legal limit (d) A exposure duration

Q9. Introgression is:

- (a) First-generation hybridisation only (b) Stable incorporation of genes via backcrossing (c) Gene transfer between microbes (d) Seed dispersal

Q10. The high-dose/refuge strategy aims to:

- (a) Eliminate all pests (b) Slow resistance evolution by maintaining susceptible alleles (c) Increase toxin dose (d) Replace monitoring

Q11. Pakistan's biosafety approvals are coordinated by:

- (a) FDA (b) National Biosafety Committee (c) EFSA (d) WHO

Q12. Which document type from the OECD summarizes crop biology (outcrossing, relatives, weediness)?

- (a) Consensus Documents (b) Red Books (c) Annex III (d) Dossiers

Section B — Understanding (Q13–Q24)

Q13. Why does a laboratory finding of non-target harm not establish field risk?

(a) Labs use wrong species (b) Because exposure in the field is usually far lower and pathway-dependent (c) Lab results are never valid (d) Because regulators ignore them

Q14. Exposure assessment answers:

(a) Is the agent toxic? (b) Will receptors contact the agent, at what magnitude and duration? (c) Is the trait useful? (d) Which law applies?

Q15. Substantial equivalence is best described as:

(a) Proof of safety (b) A comparative starting framework for assessment (c) A ban concept (d) A statistical test

Q16. The "tail" of a gene-flow decay curve implies:

(a) Gene flow stops at field edge (b) Zero flow is unattainable; management targets thresholds (c) Buffer zones are useless (d) Wind is irrelevant

Q17. Resistance-allele frequency follows an S-curve because:

(a) Mutations accelerate (b) Selection on a rare recessive allele is weak early, strong later (c) Farmers increase doses (d) Refuges add resistance

Q18. A Wilson interval is preferred over a naive binomial interval when:

(a) n is huge (b) Proportions are extreme or n small (c) Data are continuous (d) Testing means

Q19. Compositional equivalence margins must be pre-declared because:

(a) Labs require it (b) Post-hoc margins allow verdict-shopping and destroy comparability (c) Regulators distrust statistics (d) Crops change over time

Q20. Which is an *exposure-management* rather than hazard-reduction strategy?

(a) Changing toxin mode of action (b) Containment facilities for GM fish (c) Editing out a toxic metabolite (d) Reducing protein expression

Q21. Weight-of-evidence allergenicity assessment includes:

(a) A single animal test (b) Source history, sequence homology, digestibility/stability, serum screening where relevant (c) Only marketing data (d) Taste panels

Q22. Variability differs from uncertainty in that variability:

(a) Can be eliminated by better instruments (b) Is real-world heterogeneity needing design coverage, not measurement (c) Always dominates (d) Applies only to lab data

Q23. Post-release monitoring baselines are needed to:

(a) Satisfy paperwork (b) Distinguish GMO-attributable change from background variation (c) Increase yields (d) Replace hazard identification

Q24. The monarch-butterfly episode is taught mainly as a lesson in:

(a) Butterfly ecology (b) Hazard-without-exposure reasoning and communication risk (c) Maize agronomy (d) Legal liability

Section C — Application (Q25–Q36)

Q25. A GM crop with no sexually compatible wild relatives in the region: the gene-flow hazard to wild populations is:

(a) High (b) Not applicable via that route (c) Equal to crop-to-crop (d) Unknown

Q26. Outcrossing at 1 m is 3%; the decay constant gives ~e-folding per 15 m. Rough frequency at 30 m:

(a) 3% (b) ~0.4% (c) 1.5% (d) 0%

Q27. Survival: control 0.93 (n=200), field-realistic Bt pollen 0.895. First analytic step:

(a) Conclude harm (b) Compute CI of the difference and compare with comparator (c) Publish (d) Increase dose

Q28. In a resistance recursion, raising refuge from 0% to 20% will most directly:

(a) Lower initial q (b) Increase effective susceptible mating, delaying q rise (c) Increase toxin expression (d) Eliminate monitoring needs

Q29. A compositional analyte differs by 12% (CI 6–18%) but lies within the conventional reference range. Best interim classification:

(a) Hazard confirmed (b) Not flagged alone; seek consistency/mechanism follow-up (c) Equivalent by definition (d) Withdraw the dossier

Q30. Field-deposited Bt pollen protein falls with a ~3-day half-life. A lab-dose study lasting 30 days with daily renewal is therefore:

(a) Directly field-relevant (b) A conservative hazard screen, not a field-exposure surrogate (c) Invalid (d) A monitoring design

Q31. Predator LC50 is $10^3 \times$ the target's; field dose is $10^3 \times$ below target LC50. Margin of exposure for the predator:

(a) 10 (b) 10^3 (c) 10^{-3} (d) 1

Q32. In a risk matrix, a hazard scored $L=2$, $C=5$ lands where scores multiply:

(a) 3 (b) 7 (c) 10 (d) 25

Q33. Monte Carlo ERA output spans a management threshold. The most defensible next step:

(a) Ignore (b) Measure the dominant parameter better or manage conservatively (c) Round the numbers (d) Publish the point estimate only

Q34. A stacked (Cry1 + Cry2) product's refuge requirement is usually set by:

(a) The cheapest component (b) The component with the most demanding resistance profile (c) Averaging (d) Marketing

Q35. If flowering synchrony shifts two weeks under local climate, which ERA input changes first?

(a) LC50 (b) Exposure overlap factor (c) Regulatory statute (d) Comparator identity

Q36. A 90-day rodent study on a new protein shows no effect at high dose. Correct inference:

(a) Protein is harmless in all contexts (b) No indication of toxicity at tested dose/duration; integrate with other evidence lines (c) Chronic risk excluded (d) Allergenicity excluded

Section D — Analysis (Q37–Q48)

Q37. Two assessors place the same hazard one matrix-cell apart. Root cause is usually:

(a) Corruption (b) Ambiguous scale anchors (c) Different crop (d) Different laws

Q38. An equivalence test at $\pm 20\%$ passes but at $\pm 10\%$ fails. The proper conclusion:

(a) Unsafe (b) Verdict depends on the pre-declared margin; report both with justification (c) Math error (d) Comparator wrong

Q39. HGT-to-soil-microbes ranks low on consequence in ERAs despite public salience because:

(a) Bacteria avoid plants (b) Natural transformation events are exceedingly rare and establishment undemonstrated (c) It is illegal to study (d) Soil has no DNA

Q40. Stochastic resistance modelling shows 5% of replicate runs reach resistance in 10 years at 10% refuge. The management sentence should state:

(a) Resistance impossible (b) A distribution with quantified exceedance probability, not a single trajectory (c) Refuge useless (d) Model wrong

Q41. Comparing non-target survival against *unsprayed* isoline rather than the *insecticide regime replaced* biases the ERA:

(a) Against the GM option (b) Toward the GM option (c) Not at all (d) Toward organics

Q42. In TOST logic, "equivalent" requires:

(a) A non-significant t-test (b) The CI of the difference to lie wholly within margins (c) Zero difference (d) Huge n

Q43. Gene-flow monitoring detects hybrids at 150 m — beyond the modeled tail. Best response:

(a) Delete the datum (b) Re-examine model/parameters and consider long-distance mechanisms (c) Ignore — rare (d) Ban the crop

Q44. A Monte Carlo ranks flowering-overlap as the top sensitivity. Agronomic management implications:

(a) Nothing actionable (b) Planting-date/variety choices that reduce overlap become priority levers (c) Increase refuges (d) Add buffer zones only

Q45. Which finding most weakens a "negligible risk to pollinators" conclusion?

(a) High LC50 (b) Evidence larvae/pollen foraging overlaps flowering with exposure at effective doses (c) Low slope (d) Lab-only data

Q46. A dossier's non-target section reports 47 endpoints, one at $p=0.03$. Multiplicity-aware interpretation:

(a) Proven harm (b) Expected false-positive rate under $\alpha=0.05$; look for pattern/consistency (c) Fraud (d) Statistical error

Q47. Refuges fail locally while regional resistance stays low. Systems insight:

(a) Refuges irrelevant (b) Resistance dynamics are spatially structured; compliance heterogeneity matters at landscape scale (c) Monitoring failed (d) Genes drifted

Q48. Which statement keeps assessment/decision boundary intact?

(a) "Assessors must recommend approval" (b) "Assessment characterizes risk and uncertainty; regulators weigh acceptability" (c) "Public input edits hazard tables" (d) "Science decides values"

Section E — Evaluation (Q49–Q60)

Q49. Best justification for retaining substantial equivalence in teaching:

(a) It is legally binding everywhere (b) As a historical/comparative framework whose limits students must critique (c) It replaces toxicology (d) It ends debate

Q50. "Approved with conditions" is best evaluated as:

(a) Regulatory indecision (b) Risk management instruments attached to an acceptable-with-measures characterization (c) A safety denial (d) Marketing

Q51. The deepest critique of a purely hazard-based GMO law would note:

(a) Hazards are imaginary (b) It confounds hazard with risk, ignoring exposure and context (c) It is too quantitative (d) It aligns with Codex

Q52. A country approves cultivation while a trading partner approves only import. The primary scientific issue in the gap:

(a) None — thresholds/uses/exposure contexts differ by jurisdiction (b) One is wrong (c) Science differs (d) Both fraudulent

Q53. Evaluating a gene-drive ERA proposal, the controlling question is:

(a) Efficacy (b) Reversibility and containment of self-propagation (c) Cost (d) Public opinion

Q54. Which monitoring design best supports adaptive management?

(a) End-of-trial survey (b) Pre-agreed indicators + trigger thresholds + response owners (c) Annual reports without thresholds (d) Complaint-driven only

Q55. For communicating a wide uncertainty interval to the public, the best practice is:

(a) Report the point estimate only (b) State the interval, its drivers, and what would narrow it (c) Avoid numbers (d) Report worst case as certain

Q56. The strongest evidence that a compositional difference is biologically *unintended* rather than noise:

(a) Single-site significance (b) Consistency across sites/years plus mechanistic plausibility (c) Large decimals (d) Press coverage

Q57. A capstone risk assessment concluding "safe" with no uncertainty statement should be graded:

(a) Full marks (b) Failing the characterization requirement — conclusions must carry evidence + uncertainty (c) Bonus (d) Reject topic

Q58. When stacked-trait interaction testing is underpowered in field designs, the intellectually honest ERA:

(a) Claims no interaction (b) States the power limitation and applies conservative assumptions/monitoring (c) Deletes the endpoint (d) Extends patent

Q59. Which course motto best inoculates against advocacy drift?

(a) "GM is progress" (b) Trait + organism + environment + exposure + evidence + uncertainty (c) "Natural is safe" (d) "Follow the market"

Q60. The single most transferable skill from this course is:

(a) Plasmid maps (b) Structured reasoning from evidence to characterized, managed, communicated risk (c) Bacterial culturing (d) Gel electrophoresis

Answer key: [\[Answer-Key.md\]\(Answer-Key.md\)](#) · Related: [\[Short Questions\]\(Short-Questions.md\)](#) · [\[Long Questions\]\(Long-Questions.md\)](#) · [\[Viva Questions\]\(Viva-Questions.md\)](#)

Short Questions — GMO Environment and Risk Assessments

30 questions with model answers (3–5 sentence targets) · Bloom's progression: recall → evaluation

Instructor: Dr. Saira Azam, CEMB, University of the Punjab, Lahore.

SQ1. Define environmental risk assessment of GMOs.

The science-based process of identifying potential hazards of a GMO, assessing exposure of receptors in the receiving environment, characterizing risk (hazard × exposure × consequence, with uncertainty), and communicating results to inform management and regulatory decisions.

SQ2. Distinguish hazard from risk with one example.

Hazard is inherent harm potential; risk incorporates exposure and consequence. Bt toxin is a hazard to susceptible lepidopteran larvae; the *risk* to monarch butterflies depends on pollen densities on milkweed at larval season — quantified as negligible at landscape scale.

SQ3. What is problem formulation and why does it come first?

The scoping stage defining protection goals, assessment endpoints, the receiving environment, plausible hazard pathways, and the analysis plan. It precedes data work because scope determines which evidence is relevant; without it, assessment becomes unfocused hazard-listing.

SQ4. List four exposure pathways for Bt proteins from a Bt crop.

Pollen to non-target plant surfaces and larvae feeding there; plant tissue to target/non-target herbivores and onward to predators/parasitoids (tritrophic); root exudates and residue to soil organisms; crop debris to aquatic systems adjacent to fields.

SQ5. What is introgression and what four conditions does it require?

Stable incorporation of a transgene into a wild population via repeated backcrossing. Requires: sexual compatibility with a wild relative in the region; flowering overlap and shared pollinators; viable, fertile hybrids; and persistence (neutral or advantaged) of the trait in the wild background.

SQ6. Why can buffer zones not eliminate gene flow?

Pollen-dispersal curves decay steeply but retain a long-distance tail from turbulence, insect vectors, and eddies; plus gene flow also travels by seed (spillage, machinery). Buffers act on the steep segment, reducing frequency below thresholds — not to zero.

SQ7. Distinguish persistence, weediness, and invasiveness.

Persistence: a population maintaining itself (e.g., volunteers from seed bank). Weediness: persistence plus interference with agriculture (control cost, yield loss). Invasiveness: spread into natural communities beyond habitats of origin with ecological impact. A trait may affect one without the others; each has distinct evidence requirements.

SQ8. What are volunteer plants and why do they matter for HT crops?

Volunteers are plants germinating from seed shed by a previous crop. For herbicide-tolerant crops, volunteers carrying the tolerance trait narrow herbicide options in following crops — a management (not primarily ecological) risk managed by rotations, tillage, and alternative chemistry.

SQ9. State the high-dose/refuge strategy and its two conditions.

Express toxin at doses killing heterozygous-resistant insects (making functional resistance recessive), while structured refuges of non-Bt hosts maintain susceptible individuals that mate with any resistant survivors, diluting resistance alleles when rare. Conditions: genuinely high dose against heterozygotes; refuge compliance producing in-space/in-time susceptible mating.

Q10. Why is resistance monitoring described as "load-bearing" for Bt sustainability?

The refuge strategy works only while resistance alleles are rare; monitoring (diagnostic-dose bioassays, damage reports, F2 screens) is the only way to know the strategy still holds, and pre-agreed trigger thresholds convert monitoring into adaptive management before field failures.

SQ11. What does "no single test establishes allergenic safety" mean in practice?

Allergenicity is assessed as weight of evidence: source-organism history, sequence homology to known allergens, pepsin resistance/protein stability, glycosylation, serum-screening where exposure history exists. Concordant non-concern across lines supports safety; any positive flag demands targeted follow-up — no one assay is dispositive.

SQ12. What is substantial equivalence and state one limitation.

A comparative assessment framework (OECD, 1993): identify a conventional comparator and characterize differences. Limitation: it structures the question but does not prove safety — comparator choice, margin setting, and unmeasured compounds require judgment; it is a starting point, not a verdict.

SQ13. Define unintended effects and name three mechanisms.

Biological changes beyond the intended trait. Mechanisms: position effects (insertion disrupting/being regulated by local chromatin), pleiotropy (one enzyme shifting multiple metabolites), and somaclonal/insertional variation; environmental interaction can also reveal latent phenotypes.

SQ14. Why does "difference $\neq$ harm" structure unintended-effect analysis?

Detection obliges a chain: is the difference biologically relevant? adverse at real exposure? within natural ranges? Many detected differences (statistical or metabolic) are benign variation; the analysis determines which differences are adverse — not whether difference exists.

SQ15. List four uncertainty types with a one-line example each.

Measurement (LC50 CI), model (exponential vs power-law decay), sampling (rare-event detection limits), extrapolation (lab-to-field, species-to-species); plus biological variability (site-year heterogeneity) often treated separately.

SQ16. What is the purpose of a risk matrix and its key limitation?

It ranks characterized hazards (likelihood $\times$ consequence) to prioritise management and communicate reasoning. Limitation: it is presentation, not measurement — placements depend on pre-declared anchor descriptors, and false precision lurks in multiplied ordinals.

SQ17. How do Monte Carlo and sensitivity analysis complement each other?

Monte Carlo propagates input uncertainty into an output distribution (the honest interval); sensitivity analysis (rank correlations across draws) identifies which inputs drive it — telling assessors what to measure next and where conservatism buys the most.

SQ18. Distinguish risk management from risk communication.

Management selects and implements actions (conditions, refuges, containment, monitoring) based on characterization; communication is the interactive exchange of that information with stakeholders — accurate, uncertainty-explicit, neither sensational nor falsely reassuring.

SQ19. What is the Advance Informed Agreement procedure?

Cartagena Protocol requirement that an exporter/part provide information and obtain the importing party's consent before the first intentional transboundary movement of an LMO for environmental introduction — making prior risk assessment and notification an international obligation.

SQ20. Why does the EU system require PMEM while regulatory practice elsewhere varies?

EU regulation (2001/18) ties time-limited authorisations to mandatory monitoring for unanticipated effects, embedding adaptive review; other systems rely more on case-specific conditions. It reflects a process-based, precaution-oriented regulatory philosophy rather than a scientific difference in what can be monitored.

SQ21. What should a post-release monitoring baseline contain?

Pre-release reference data for chosen indicators (target/non-target populations, resistance frequencies, wild-relative hybridization rates) across representative sites and multiple seasons, with sampling design matched to the power needed to detect the trigger thresholds later agreed.

SQ22. Give one good-practice and one bad-practice example of risk communication.

Good: "Under these assumptions, exposure stays $\geq 100\times$ below the laboratory LC50; the margin's dominant uncertainty is pollen-deposition measurement, which we propose to refine." Bad: "GM pollen kills butterflies" — a hazard fragment presented as a risk conclusion.

SQ23. How does the UK Farm-Scale Evaluations lesson generalise?

Field-scale ecological effects tracked the *herbicide-management regime*, not the transgene alone — different HT systems gave opposite community outcomes. Generalisation: assess management systems and their realistic alternatives, not traits in isolation.

SQ24. Why is comparator choice called "the most consequential early decision" in an ERA?

Comparators define "different" — sprayed-conventional vs unsprayed-isoline baselines reverse non-target conclusions; near-isogenic vs commercial-reference sets compositional flags. The comparison, not the arithmetic, determines most downstream characterizations.

SQ25. What makes GMM (microorganism) ERAs structurally different?

Survival, dispersal, and horizontal gene transfer potential dominate; populations are vast, mobile, and exchange genes — so reversibility, containment, and survival/dispersal data replace the plant ERA's focus on field demography and outcrossing.

SQ26. What is tiered testing and its escalation criterion?

Lab maximum-hazard testing → semi-field → field, escalating only when a lower tier shows concern at *realistic exposure*; the criterion prevents both wasted field trials on trivial hazards and premature "no risk" claims from weak designs.

SQ27. State one strength and one weakness of the US Coordinated Framework.

Strength: product/use trigger leverages existing, expert statutes (pesticide, plant-pest, food law) efficiently. Weakness: process-blindness can leave novel products with no clear home (gaps historically acknowledged), and framework coherence depends on interagency coordination.

SQ28. What is the difference between general surveillance and case-specific monitoring?

Case-specific: targeted to identified hazards from the ERA (e.g., resistance bioassays around specific refuges). General: broad surveys for unanticipated effects across systems. Both feed PMEM; triggers and responsibilities should be pre-agreed for each.

SQ29. Why is "acceptable risk" a policy concept rather than a scientific output?

Science characterizes risk magnitude, probability, and uncertainty; whether that residual is tolerable given benefits, alternatives, and societal values is a decision norm — jurisdictions with identical assessments may legitimately land differently.

SQ30. What three things must every student conclusion in this course carry?

An evidence base with named comparators; an uncertainty statement (interval or flagged gaps, not adjectives); and a management/monitoring instrument with a trigger — assessment without instruments and honesty about limits is incomplete.

Related: [MCQs](MCQs.md) · [Long Questions](Long-Questions.md) · [Viva Questions](Viva-Questions.md) · [Answer-Key](Answer-Key.md)

Long Questions — GMO Environment and Risk Assessments

20 essay questions with structured answer outlines · Analysis → Evaluation → Synthesis

Instructor: Dr. Saira Azam, CEMB, University of the Punjab, Lahore.

LQ1. Explain the complete GMO environmental risk-assessment framework, stage by stage.

Outline: problem formulation (protection goals, endpoints, environment) → hazard identification (agent–mechanism–endpoint) → hazard characterization (dose–response, tiers) → exposure assessment (pathways, magnitude/duration) → risk characterization (integration) → uncertainty analysis → management → monitoring → communication. For each: inputs, questions, outputs, limitations; close with one worked example threaded through all stages (Bt cotton recommended).

LQ2. "Hazard ≠ Risk." Discuss with two contrasting GMO examples.

Outline: definitions; conceptual equation and its non-universality; Case 2 (monarch — hazard without exposure → negligible risk) vs Case 1/3 (resistance/herbicide regimes — hazard realized *through management systems*); implications: tiered testing, exposure quantification as decisive, communication discipline.

LQ3. Describe the biology and assessment of pollen-mediated gene flow, and derive management implications.

Outline: dispersal biology; crop-to-crop vs crop-to-wild; decay model $f(d)=a \cdot e^{(-bd)}$ with threshold logic (Lab 03); introgression conditions (SQ5); evidence: OECD biology dossiers, hybridization/fitness studies; management: isolation, borders, timing, hygiene; residual tail → acceptability as policy.

LQ4. Compare persistence, weediness, and invasiveness, and construct the assessment workflow for a proposed GM crop.

Outline: definitions and differentia (SQ7); trait characterization → phenotypic comparison → environment → reproductive fitness → persistence potential → consequence; how GM traits might alter each (dormancy, stress tolerance, seed output); evidence: fitness-component trials, seed-bank studies; conclude: trait × environment specificity.

LQ5. Critically evaluate tiered non-target testing using Bt crops as the case.

Outline: tier logic and escalation criteria (SQ26); lab maximum-hazard → semi-field → field; monarch case as exposure-quantification exemplar; tritrophic routes (Lab 04); comparator selection bias (Q41); limits: sublethal/chronic endpoints, community effects; weight-of-evidence conclusion.

LQ6. Explain resistance evolution to Bt crops and evaluate the refuge strategy's successes and fragilities.

Outline: selection genetics (S-curve, recessive rarity, Lab 05 recursion); high-dose/refuge conditions; pyramids and cross-resistance; monitoring bioassays and triggers; fragilities: compliance, $q_{\square}$ uncertainty, fitness costs; herbicide-resistance parallel (system-level selection, Module 12).

LQ7. Discuss soil-ecosystem assessment of GM crops: exposure, methods, and interpretation discipline.

Outline: exposure routes (exudates, residue, DNA/protein persistence); methods (community profiling, functional assays, decomposition); findings generally small/transient but study-dependent; interpretation

chain difference→relevance→adverse; natural spatial/temporal variability as the dominant noise source; monitoring integration.

LQ8. Explain how effects propagate through food webs and why single-species evidence under-determines ecosystem risk.

Outline: trophic structure; direct vs indirect/trophic effects; buffering/amplification in networks; UK FSE as regime-level exemplar (Case 15); conceptual and quantitative web models (Module 25.4); why community conclusions lag; indicators and their baselines.

LQ9. Compositional assessment: design, statistics, and what it can and cannot establish.

Outline: comparator/reference-range design (Lab 07); TOST/equivalence vs t-test; pre-declared margins; multiplicity; intended vs unintended differences; limits: unmeasured compounds, processing, interactions — one evidence line in weight-of-evidence (Module 14).

LQ10. Allergenicity assessment: present the weight-of-evidence framework and justify why no single test suffices.

Outline: each line (source history, homology, pepsin stability, glycosylation, serum screening) with what it can/cannot detect; discordant-evidence handling; StarLink case as regulatory consequence of unresolved allergenicity (Case 5); honest reporting of residual uncertainty.

LQ11. Molecular characterization: what does the ERA need and why?

Outline: construct/insert characterization (copy number, site, integrity, stability), expression profiling, protein identity — links to hazard (what is expressed, where, how much) and to unintended-effect screening; genome-edited organisms: intended + potential off-target edits, comparability to natural variation; regulatory divergence implications (Module 22, 25.8).

LQ12. "Difference is not automatically harm." Construct the full interpretive chain and apply it twice.

Outline: observed difference → biological relevance → adverse? → exposure? → risk?; applications: a soil-community shift (Module 9) and a compositional analyte (Lab 07); role of natural ranges, consistency, mechanism; conclusion: the chain as the course's interpretive core.

LQ13. Discuss uncertainty in ERA: taxonomy, quantification, and communication.

Outline: five types (SQ15); quantification: CIs, Monte Carlo, model ensembles; variability vs uncertainty separation (Lab 10); uncertainty matrix; communication: intervals, drivers, what would narrow them; precaution as policy response, not data substitution.

LQ14. Risk characterization: integrate evidence into a defensible conclusion for one GMO of your choice.

Outline: hazard evidence + exposure + dose–response + ecological context + uncertainty (Module 17); qualitative vs quantitative modes; matrix placement with pre-declared anchors (Lab 09); the conclusion's required parts (SQ30); chosen-GMO example (capstone preview).

LQ15. Evaluate risk-management instruments for agricultural GM crops and match each to a characterized risk.

Outline: instrument classes: avoidance/containment, mitigation (refuges, rotations, mixtures), buffer zones, stewardship, monitoring with triggers; mapping table risk→instrument→trigger→owner (Module 18); compliance as binding constraint (Bt cotton); containment substitution for exposure (salmon case).

LQ16. Design a post-release monitoring program for a specified GM crop in Pakistan.

Outline: baseline requirements (SQ21); indicators: resistance frequencies, gene-flow sentinels, non-target assemblages; trigger thresholds and responses (Case 13 logic); adaptive management loop (Module 19); constraints: resources, compliance, climate variability; reporting obligations under national rules.

LQ17. Compare the US, EU, and Canadian regulatory approaches and analyze why the same event can face different outcomes.

Outline: product/use (Coordinated Framework) vs process (2001/18) vs novel trait; assessment/decision split; implications for gene-edited crops (25.8, Case 11); trade and coexistence consequences; evaluation: coherence, gaps, incentives.

LQ18. Present Pakistan's biosafety framework and critically assess its implementation capacity.

Outline: PEPA 1997 + Biosafety Rules 2005; NBC/TAC/IBC structure; Bt cotton approval history (verify current notifications); capacity: lab infrastructure, monitoring enforcement, compliance incentives; Cartagena obligations; balanced conclusions on strengths/gaps — with authoritative-source citations.

LQ19. Discuss risk communication failures and successes using two course cases; derive best-practice principles.

Outline: monarch (hazard headline preceding exposure data) vs post-quantification correction; StarLink (use-specific approval colliding with supply reality); principles: hazard/risk precision, comparators, intervals, admitting unknowns without false reassurance (Module 21).

LQ20. Advanced frontier essay: which emerging technique (gene drive, gene-edited wild-relative engineering, or synthetic-bio release) most challenges existing ERA frameworks, and why?

Outline: framework transferability (Module 25); self-propagation → reversibility dominance; introgression-rate assumptions broken (drives); molecular-characterization burden for edits; containment logic for synthetics; regulatory readiness gaps; structured, evidence-anchored argument, no advocacy.

Related: [MCQs](MCQs.md) · [Short Questions](Short-Questions.md) · [Case-Based Questions](Case-Based-Questions.md) · [Answer-Key](Answer-Key.md)

Viva Questions — GMO Environment and Risk Assessments

40 rapid-response questions with model answers · oral-examination format

Instructor: Dr. Saira Azam, CEMB, University of the Punjab, Lahore.

Basics (Q1–Q8)

Q1. Define risk assessment in one sentence.

Characterizing, with stated uncertainty, what harm a GMO could cause, to what, how likely — to inform management and decisions.

Q2. Give the four components of the conceptual risk relationship.

Hazard, exposure, dose–response, context (conceptually combining to risk).

Q3. What is the receiving environment?

The ecosystem/agroecosystem into which the GMO would be released — its species, climate, practices — defining exposure contexts.

Q4. Name the ERA stages in order.

Problem formulation → hazard identification → hazard characterization → exposure assessment → risk characterization → uncertainty → management → monitoring → communication.

Q5. What is an assessment endpoint?

The valued ecological property to protect (e.g., pollinator abundance, soil function) that measurement endpoints proxy.

Q6. What does case-by-case mean here?

Each GMO × environment combination is assessed on its own evidence — no blanket judgments across traits or organisms.

Q7. What is a protection goal?

What society/regulation wants kept unharmed (biodiversity, ecosystem services) — set in problem formulation, often policy-derived.

Q8. One-line: why assess at all when harm isn't proven?

To decide knowingly: quantify what could go wrong, how likely, how serious — before, not after, release.

Concepts (Q9–Q16)

Q9. Hazard vs risk — with the glass-cobra image.

Hazard = venomous bite potential; risk = it behind thick glass = near zero. Exposure gates the outcome.

Q10. Why isn't "no evidence of harm" proof of safety?

It may mean no competent search — power, endpoints, and duration determine what absence of evidence is worth.

Q11. What makes a hazard statement "assessable"?

Agent + mechanism + endpoint specified — data can then support or refute it.

Q12. Difference between measurement uncertainty and variability?

Measurement shrinks with better instruments; variability is real-world heterogeneity needing design coverage (more sites/years).

Q13. What is a comparator and why consequential?

The reference (near-isogenic, conventional regime) against which difference is judged — defines "different," steering all conclusions.

Q14. Weight of evidence — two sentences.

Integrate multiple imperfect lines; conclusions strengthen with convergence, weaken with unexplained conflict.

Q15. What is "familiarity"?

OECD concept: prior knowledge of organism biology and use reduces novelty-driven uncertainty — it informs, never replaces, assessment.

Q16. Define exposure pathway.

Route linking source (GMO/protein) to receptor: e.g., pollen → deposition → non-target larva → predator.

Ecology (Q17–Q24)

Q17. Name three non-target functional groups and one exposure route each.

Pollinators (pollen/nectar), predators (prey-borne protein), soil Collembola (residue/exudates).

Q18. What is a tritrophic route?

Plant → herbivore → natural enemy; e.g., parasitoid developing inside Bt-intoxicated hosts.

Q19. Why do tier-1 tests use maximum hazard?

Screening conservatively: if even forced dose shows nothing, escalation stops; if effect appears, realistic-exposure tiers decide.

Q20. What did the monarch case quantify?

Field pollen densities on milkweed, per-event toxicity, phenology overlap — all far below lab-effect doses → negligible landscape risk.

Q21. Two structural differences of GMM ERAs?

Survival/dispersal dominance and HGT potential; reversibility/containment central.

Q22. What is introgression's minimal condition set?

Compatible relative + hybrid viability/fertility + backcross persistence (neutral or advantaged).

Q23. Why is the herbicide regime, not the HT trait, the usual resistance driver?

Selection acts where the pressure is — repeated glyphosate on weeds; the trait enabled the regime, the regime imposed the selection.

Q24. What is an ecosystem-service framing of risk?

Assessing harm to services (pollination, biological control, nutrient cycling) — tying endpoints to societal protection goals.

Quantitative (Q25–Q32)

Q25. Why Wilson over Wald intervals?

Better small-n/extreme-proportion behaviour — standard in survival/hybridization counts.

Q26. Interpret $f(d)=a \cdot e^{(-bd)}$.

Edge frequency a decaying at rate b ; threshold distance $d^* = \ln(a/\text{threshold})/b$ for management design.

Q27. What does an S-shaped resistance curve reveal?

Selection on rare recessives is weak early, compounding later — refuges exploit exactly that window.

Q28. LC50 vs NOEC — one sentence each.

LC50: dose killing half the cohort (hazard parameter). NOEC: highest tested concentration without observed effect (design-limited, falling from regulatory favour).

Q29. Margin of exposure formula and use.

MOE = (effect dose, e.g., LC50) / (estimated field exposure) — ratios $\gg 1$ support negligible-risk statements, with uncertainty.

Q30. What is TOST?

Two one-sided tests proving the difference CI lies within pre-declared equivalence margins — the correct "equivalence" tool.

Q31. Monte Carlo output is a distribution. What's reported?

Median plus interval (e.g., 90%) and exceedance probabilities against thresholds — never just the mean.

Q32. What does a tornado plot show?

Output swing as each input moves low→high bound — visual sensitivity ranking.

Regulation & ethics (Q33–Q40)

Q33. Cartagena Annex III in one sentence.

Principles for LMO risk assessment: case-by-case, science-based, step-wise, transparent, with precaution considerations.

Q34. Which three US agencies and their domains?

USDA-APHIS (plant pest), EPA (PIPs/pesticides), FDA (food/feed).

Q35. EU process-trigger vs Canada product-trigger — one example each of divergence.

EU: gene-edited small deletions captured as GMOs. Canada: a conventionally-bred novel trait captured while the edit might not be.

Q36. Pakistan's three committee tiers?

NBC (approvals/policy), TAC (technical review), IBCs (institutional oversight).

Q37. PMEM vs general surveillance?

Case-specific monitoring targets ERA-identified hazards; general surveillance watches broadly for the unanticipated — both under pre-agreed triggers.

Q38. Why keep benefits out of the safety characterization?

Conflating them smuggles policy into hazard science; benefits belong to the separate decision layer.

Q39. One mark of advocacy drift in a student report?

Conclusions outrunning evidence in either direction — e.g., "safe" without uncertainty, or "dangerous" without exposure.

Q40. The course motto?

Trait + organism + environment + exposure + evidence + uncertainty — every GMO evaluated as its own case.

Related: [MCQs](MCQs.md) · [Short Questions](Short-Questions.md) · [Long Questions](Long-Questions.md) · [Answer-Key](Answer-Key.md)

Case-Based Questions — GMO Environment and Risk Assessments

15 scenario questions with guided answers · Application → Evaluation

Instructor: Dr. Saira Azam, CEMB, University of the Punjab, Lahore.

CQ1. A company proposes Bt chickpea for Pakistan, where wild *Cicer* relatives occur in some districts. Structure the problem formulation.

Guide: protection goals (wild *Cicer* genetic integrity, non-target lepidopterans, farm sustainability); hazards to scope: crop-to-wild gene flow (Module 6), non-target exposure via pollen/tissue (Module 8), resistance in pod-borer (Module 11); environment specification: districts with relatives, cropping calendar, pollinator fauna; analysis plan mapping each hazard to required data.

CQ2. A lab study shows 25% mortality of a beneficial lacewing fed Bt-treated eggs at 100× field dose. The developer claims "no risk." Assess the claim.

Guide: claim overreaches — tier-1 hazard finding at forced dose; correct next step: realistic-exposure semi-field/field tiers, tritrophic route analysis (does the lacewing consume intoxicated prey?), comparator (insecticide regime replaced). "No risk" is unproven *and* unproven danger — the honest state is "hazard at high dose; exposure-dependent risk under quantification."

CQ3. Monitoring finds transgene hybrids in a wild relative at 2 km from the nearest GM field — beyond the modelled tail. What follows?

Guide: do not discard the datum; re-examine model structure (long-distance mechanisms: turbulence, insect vectors, unrecorded fields, seed movement); re-estimate parameters; check consequence (fitness effect of trait in recipient — Lab 03/06 logic); update management if threshold crossed; report model uncertainty honestly (Module 16).

CQ4. A HT rice line escapes herbicide regimes; volunteers persist three seasons. Characterize the risk chain.

Guide: trait (tolerance) × organism (rice volunteers) × environment (rotations) → exposure (volunteer presence in follow-up crops) → consequence (narrowed herbicide options, seed-bank persistence — Module 7); management: rotation design, alternative chemistry, machinery hygiene; monitoring: volunteer counts with triggers.

CQ5. A stack (Cry1F + Cry3Bb + HT) shows no single-protein interaction in lab mixes but field refuges are not enforced. Evaluate the resistance risk profile.

Guide: components individually characterized; interaction low under tested conditions — state power limits (Q58 logic); dominant risk is compliance-dependent refuge failure (Case 1 lesson); risk statement must carry compliance as an exposure parameter; instruments: enforcement-linked conditions, monitoring triggers, stewardship audits — not more lab data.

CQ6. An assessor's Monte Carlo output: median flow 0.12%, 90% interval 0.03–0.45%; threshold 0.1%. Draft the risk statement.

Guide: template (Lab 10): state median/interval, dominance (decay parameter), exceedance share, response options (measure b better / manage conservatively / robust design); explicitly refuse a bare "safe/unsafe" verdict; connect to buffer-distance choice.

CQ7. Regulators receive two compositional datasets: 50 analytes, one at $p=0.02$ (fat, +4%, within conventional range). The NGO demands rejection. Advise.

Guide: multiplicity (expected ~2–3 false positives at $\alpha=0.05$ — Lab 07 challenge); within-range + tiny magnitude + single test → flag for pattern analysis, not rejection; check consistency across sites; the decision layer may still consider it — but the assessment's characterization is "no indication of unintended

compositional change" pending consistency data.

CQ8. A GMM producing a biocontrol enzyme is proposed for open-field spray. List the assessment elements that differ from a crop ERA.

Guide: survival/dispersal kinetics, population establishment/reversibility, HGT potential (Module 9/25.9 logic), containment contingency, environmental fate of cells vs proteins; monitoring: viability persistence, gene-transfer sentinels; management: application windows, buffer zones, kill-switch/auxotrophy engineering if claimed — verify.

CQ9. A gene-edited (SDN-1) herbicide-tolerance crop faces different regulatory tracks in two markets. Frame the scientific vs regulatory components.

Guide: scientific: molecular characterization (edit confirmation, off-target screen), phenotypic/compositional equivalence — the same either way; regulatory: trigger (process vs product) differs; assessors can characterize identically while regulators decide differently (Module 22) — separating the layers is the answer.

CQ10. Post-release year 3: resistance bioassays show 4% survivors at diagnostic dose (trigger = 2%). What does adaptive management require?

Guide: trigger fired → pre-agreed response, not ad-hoc debate: verify assay, increase refuge/enforcement, rotate/stack chemistry or toxin, intensify monitoring frequency, alert stewardship network (Module 19; Case 13); document decision trail; revise ERA assumption (q■/dominance) with field data.

CQ11. A ministry official asks: "Is Bt cotton safe?" Draft the two-sentence scientifically defensible reply.

Guide: "Assessments to date, across jurisdictions, find the characterized risks acceptable under specified conditions — chiefly effective refuges and monitoring; the dominant documented risk is resistance evolution where management lapses." Then: "No assessment proves absolute safety; it quantifies risk and uncertainty, and conditions sustain acceptability." (Module 21 discipline.)

CQ12. An NGO publishes a map of "contamination" incidents (gene flow detections) implying ecological harm. Critique.

Guide: detections ≠ harm (Q25 logic): consequence depends on trait fitness effect in recipient (often neutral); distinguish crop-to-crop commingling (coexistence issue) from crop-to-wild introgression (ecological issue); demand the exposure→consequence chain before harm claims — while acknowledging transparency value of the underlying data.

CQ13. A developer's dossier asserts "no unintended effects" citing only compositional equivalence. Identify the gaps.

Guide: unintended-effect analysis requires the interpretive chain (Module 16): molecular characterization (position/pleiotropy screen), phenotypic/agronomic comparison across environments, expression profiling, and honest treatment of detected-but-benign differences; "no unintended effects" is unfalsifiable as phrased — require "no indication of unintended change beyond characterized differences."

CQ14. Two assessors score volunteer-persistence L=3,C=3 and L=4,C=2 — same cell, same product. What does this reveal and what is fixed?

Guide: identical product scores → anchors not scorers are the issue (Lab 09); fix: frequency-anchored likelihood descriptors and magnitude-anchored consequence descriptors, pre-declared; document; re-score — the matrix improves by descriptor iteration, not scorer calibration.

CQ15. Design the executive-summary risk paragraph (5 sentences) for a proposed GM soybean in a region with no wild relatives and glyphosate-resistant weeds already present.

Guide: (1) trait/organism/environment specificity; (2) gene-flow risk negligible via compatible-relative route (none present), crop-to-crop manageable by segregation; (3) dominant characterized risk: herbicide-regime resistance selection — already present locally, trait could intensify; (4) instruments: mandated rotation/mixture stewardship + resistance monitoring triggers; (5) uncertainty: volunteer dynamics across rotations — flagged with proposed measurement. Full-capstone logic in miniature.

Related: [Case Studies module](../MARKDOWN/23-Case-Studies.md) · [MCQs](MCQs.md) · [Answer-Key](Answer-Key.md)

Data-Interpretation Questions — GMO Environment and Risk Assessments

15 questions on the course's simulated datasets · each: read → compute → interpret → limit

Instructor: Dr. Saira Azam, CEMB, University of the Punjab, Lahore. Datasets: [DATA]/(../DATA/); answers reproducible via `DATA/analysis\_demo.py`.

DQ1. [gene-flow] From `gene_flow_by_distance.csv`, report mean frequency at each distance and state the direction/shape of decay. What single sentence conveys the management implication?

Guide: steep exponential-style decay from $\approx 3\%$ (1 m) toward $\approx 0.02\%$ (100 m); sentence: "buffer distances act on the steep segment; the tail persists, so management targets thresholds, not zero."

DQ2. [gene-flow] Fit $\ln(\text{frequency})$ vs distance; report a , b , R^2 . At what distance does the point estimate cross 0.001? Why is the upper-bound distance larger?

Guide: $\ln$ -linear fit; $d^* = \ln(a/0.001)/b$ ($\approx 15\text{--}30$ m in the simulated data); the 95% bound sits above the point estimate, so a confidence-based buffer is longer — regulator-relevant distinction.

DQ3. [non-target] Compute survival proportions and Wilson 95% CIs for all treatments. Which comparisons are compatible with no effect?

Guide: field-realistic Bt-pollen vs control overlapping CIs → compatible with no effect; high-dose may show a small significant drop (hazard flag); conventional-insecticide group markedly lower (comparator realism).

DQ4. [non-target] A colleague reports "Bt pollen significant at $p=0.04$ " for parasitoid emergence. Using the dataset, write the three-sentence correct interpretation.

Guide: effect size first (tiny), interval nearly touching null, multiplicity context; comparator (insecticide) dwarfs it; conclusion: flag for confirmation, not a risk conclusion — and no single-season claim (Lab 04 discipline).

DQ5. [resistance] Plot q vs generation for 0/10/20% refuge; report generation of $q=0.5$ each. Why is the curve flat, then steep?

Guide: S-curve — rare-recessive selection weak early ($\Delta q \propto q^2$), compounding late; refuges add susceptible mating throughout, stretching the flat phase dramatically (documented dramatic refuge effect).

DQ6. [resistance] Vary q (1e-4 → 1e-2) at 10% refuge. Which sensitivity is larger — doubling q or halving the refuge? Justify from the recursion.

Guide: typically halving refuge dominates (mating-pool dilution is the engine); q matters mostly for timing onset. Both must be reported conservatively because both are hard to measure (Module 11).

DQ7. [exposure] From pollen-decay and protein half-life data, compute the field-edge receptor dose chain and its hazard quotient at 5/25/100 m.

Guide: multiply transfer factors through the pathway (Lab 06); HQ at 5 m largest but still $\ll 1$ in simulated data; at 100 m margin widens by orders of magnitude; identify the dominant parameter (edge deposition).

DQ8. [exposure] Half-life estimate: from protein decay time series, fit exponential and report half-life with CI. Why does the CI matter more than the point?

Guide: HQ sensitivity to persistence (longer half-life → longer exposure windows); interval crossing days-to-weeks changes residue-timing guidance — point estimates invite false precision.

DQ9. [composition] Build the per-analyte table: % diff, CI, within-reference-range? Which analytes are (i) intended differences, (ii) flagged, (iii) benign?

Guide: classify with TOST at $\pm 20\%$ (Lab 07); intended: the trait's target metabolite; flagged: pattern/consistency test before calling unintended; benign: within-range single-test differences (multiplicity).

DQ10. [composition] One analyte: comparator 1.20, GM 1.34, CI of difference [0.06, 0.22], conventional range 0.9–1.9. Verdict chain?

Guide: significant, marginally outside $\pm 20\%$ at upper bound, well within natural range → not equivalent by test but biologically unremarkable; response: consistency check across sites, mechanism search; record as "within natural variation; monitored."

DQ11. [dose-response] Report LC50 and slope per species; compute the target→non-target selectivity margin. What field-exposure line would make the margin moot?

Guide: simulated margins span orders of magnitude; margin moot if field exposure approached the *non-target* LC50 — i.e., only at doses far above any documented pollen/tissue exposure; report as inequality when unmeasured.

DQ12. [dose-response] A shallow slope for the Collembolan means what for threshold-setting, given $\pm 3\times$ measurement error on concentration?

Guide: shallow β → response shifts greatly with small dose error; thresholds need uncertainty buffers (use CI, not point); ties to chronic/sublethal caveat — LC50 logic doesn't cover them (Lab 08 limits).

DQ13. [risk-matrix] Using pre-declared anchors, place eight hazards; then re-score without comparator studies. Which placements moved and why?

Guide: resistance ranks top with comparators (regime replaced is worse); non-target and protein-food hazards drop without comparators inflating them → comparator availability, not logic, shifts rankings (Lab 09 challenge).

DQ14. [uncertainty] Run the 10,000-draw MC; report median, 90% interval, and rank-correlation sensitivities. Which input dominates and what measurement campaign follows?

Guide: isolation-distance/decay parameter b dominates (documented); campaign: multi-distance pollen-decay study across seasons, prioritized over edge-frequency replication — sensitivity analysis as resource allocation.

DQ15. [uncertainty] The 90% interval straddles the 0.1% threshold. Write the regulatory sentence that neither over- nor under-states this.

Guide: "Median estimate 0.12% (90% CI 0.03–0.45%) exceeds the 0.1% threshold in ~55% of draws, driven by decay-parameter uncertainty; decision options are refined measurement of distance-decay or conservative buffer management while measured." — interval + driver + response, no adjectives (Lab 10 template).

Datasets: [DATA/README](../DATA/README.md) · *Related labs:* [Lab 03](../LAB/Lab-03-Gene-Flow-Data-Analysis.md)–[Lab 10](../LAB/Lab-10-Uncertainty-Sensitivity-Analysis.md) · [Answer-Key](Answer-Key.md)

Answer Key (Instructor Copy)

GMO Environment and Risk Assessments · Instructor: Dr. Saira Azam, CEMB, University of the Punjab, Lahore.

Covers [MCQs](MCQs.md), [Short Questions](Short-Questions.md), [Long Questions](Long-Questions.md), [Viva Questions](Viva-Questions.md), [Case-Based](Case-Based-Questions.md), [Data-Interpretation](Data-Interpretation-Questions.md). Short/Long/Viva/Case/Data answers are embedded in their files; this key gives MCQ answers plus grading rubrics.

MCQ Answer Key

Q	Ans	Why (one line)
1	c	Hazard = inherent harm potential
2	b	Characterization integrates all four
3	b	Cartagena = transboundary LMOs
4	c	Problem formulation precedes all
5	b	Post-Market Environmental Monitoring
6	c	EPA registers PIPs under FIFRA
7	b	Sexual compatibility is the gate
8	b	Lethal concentration for 50%
9	b	Backcross incorporation
10	b	Susceptible-allele dilution
11	b	National Biosafety Committee
12	a	OECD consensus (biology) documents
13	b	Field exposure far lower/pathway-bound
14	b	Exposure = contact magnitude/duration
15	b	Comparative starting framework
16	b	Tail → thresholds, not zero
17	b	Rare-recessive dynamics
18	b	Wilson for small-n/extremes
19	b	Pre-declaration prevents verdict-shopping
20	b	Containment manages exposure
21	b	WoE allergenicity lines
22	b	Variability needs design, not instruments
23	b	Baselines enable change detection
24	b	Exposure-quantification lesson
25	b	No compatible relative → route N/A
26	b	$3\% \cdot e^{(-2)} \approx 0.4\%$
27	b	CI + comparator first
28	b	Refuge raises susceptible mating
29	b	Within-range → follow-up, not flag alone
30	b	Conservative screen ≠ field surrogate
31	b	$10^2 \times 10^3 = 10^5$
32	c	$2 \times 5 = 10$

Q	Ans	Why (one line)
33	b	Measure-dominant or manage conservative
34	b	Most demanding component governs
35	b	Overlap factor is the exposure gate
36	b	One line among many; scoped inference
37	b	Anchor ambiguity is the root cause
38	b	Margin-dependent verdicts, both reported
39	b	Rarity + no demonstrated establishment
40	b	Distributions, not single trajectories
41	a	Unsprayed baseline inflates apparent harm
42	b	CI wholly inside margins
43	b	Model re-examination, not deletion
44	b	Overlap reduction = priority lever
45	b	Realistic effective exposure is decisive
46	b	Expected false positives; seek pattern
47	b	Landscape compliance heterogeneity
48	b	The assessment/decision boundary itself
49	b	Teach it to critique it
50	b	Conditions = instruments on acceptability
51	b	Hazard-law confounds hazard with risk
52	a	Thresholds/uses differ legitimately
53	b	Reversibility/containment dominate
54	b	Indicators + triggers + owners
55	b	Interval + drivers + narrowing path
56	b	Consistency + mechanism, not one p
57	b	Fails characterization requirement
58	b	Power limit stated, conservative assumptions
59	b	The course motto
60	b	Evidence → characterized, managed, communicated risk

Scoring suggestion: Section A 1 mark each; B 1.5; C 2; D 2.5; E 3 — total 110, scale to 100. Pass ≥50; distinction ≥75.

Grading rubrics (short/long/case/data answers)

Short questions (3–5 sentences): definition accuracy (40%) · example/structure (30%) · nuance — uncertainty, comparators, limits (30%). Full marks require at least one honest-limit clause.

Long questions: outline coverage (40%) · evidence anchoring (course cases/data, 25%) · interpretive discipline (difference≠harm; assessment≠decision — 20%) · structure/clarity (15%). Advocacy drift in either direction caps at 60%.

Case-based: correct chain hazard→exposure→consequence (40%) · instrument proposed (25%) · uncertainty carried (20%) · science/policy separation (15%).

Data-interpretation: computation correct (35%) · interpretation in ERA language (35%) · limitation named (20%) · units/design checks (10%).

Viva: full marks = precise one-sentence answers; half marks = correct-but-vague; zero for hazard/risk conflation or unfalsifiable claims.

Common wrong-answer patterns (for feedback)

1. "Safe/dangerous" verdicts without exposure or uncertainty → return to Module 2 chain.
2. Treating lab hazard findings as risk conclusions → monarch case re-read.
3. Significance-without-effect-size readings → Lab 04 three-sentence drill.
4. "Zero gene flow is achievable" → decay-tail discussion, Lab 03.
5. Refuges seen as compliance-optional → Case 1 + DQ6.

Related: [MCQs](MCQs.md) · [Short Questions](Short-Questions.md) · [Long Questions](Long-Questions.md) · [Viva Questions](Viva-Questions.md) · [Case-Based Questions](Case-Based-Questions.md) · [Data-Interpretation Questions](Data-Interpretation-Questions.md)